TBP03: Summary Document

Overview

This document summarizes the lecture on backtesting and live trading strategies using the Blueshift platform.

The lecture introduces the workflow of the Blueshift platform, the use of its API, implementation of demo strategies and the process to backtest and live trade through the platform.

The lecture covers the following topics:

- Basic definitions
- What is a trading strategy?
- The strategy spectrum
- What is Blueshift?
- Systematic strategy design cycle
- Trading system from user's perspective
- How to develop a systematic trading strategy?
- Blueshift workflow
- Alpha strategies: Understanding the source of risks and returns
- Systematic equities: A brief overview
- Blueshift pipelines
- Statistical arbitrage
- Systematic trading in FX
- Backtest evaluation
- Systematic strategy: What it is not?
- Recommendations for building strategies
- List of demo strategies

Basic definitions

1. **Systematic trading**: Trading decisions arrived at methodically with defined trading goals, risk controls and rules

2. **Quantitative trading**: Trading signal generation based on quantitative analysis

3. **Algorithmic trading**: Execution of trades using automated pre-programmed trading instructions

4. **High-frequency trading**: Algorithmic trading characterized by high speed, high turnover rates and high order-to-trade

There are three degrees of freedom -

- How trading and or other decisions are evaluated (quant or qualitative signal generation)?
- How trading and other decisions are finalized (rule-based or discretionary decisions/order generations)?
- How are those trades executed (automated or manual)?

What is a trading strategy?

Strategy: A set of rules or a well-defined algorithm to choose an action from collecting all possible actions in any state of a game.

Trading strategy: The game is stochastic (state transitions are probabilistic), massively multiplayer, and potentially long-running.

The game plan

1. Usually, the number of participants is large; hence we create the concept of "market" (unlike smaller games like auctions or business negotiations)

2. Essentially the market manifests itself in terms of the tradable assets (what you can trade), their prices (at what levels) and the liquidity (how much)

3. We make assumptions to understand how the state changes -
    a. A set of theories from economics that maps real-world to trading conditions and another set of theories on how the market processes information to reflect changes in those conditions

4. The first set of theories is, in general, known as **asset pricing theories.**

5. The second group of theories talks about **market efficiency** and **behavioural economics.**

The strategy spectrum

The following snapshot shows the ideas implemented through various techniques.

Quant	Momentum (time-series or cross-sectional)	Pair-trading, most types of statistical arbitrage	Advanced models (e.g. HMM, regime switching)	HF Market-making, Cash-futures arbitrage	News-based automated trading
Technical	MA cross-over, Continuation patterns	Swing, Retracement, Pivot trading	Opening range, dual thrusts, patterns	Range-based short gamma (vol selling)	Nothing much here
Fundamental	Factor-based investing	value investing	value/ RV (relative value) strategies	Cross-asset, cross country RV/ short gamma	Usually discretionary
	Trending	Mean-reverting	Break-out	Carry	Event-based

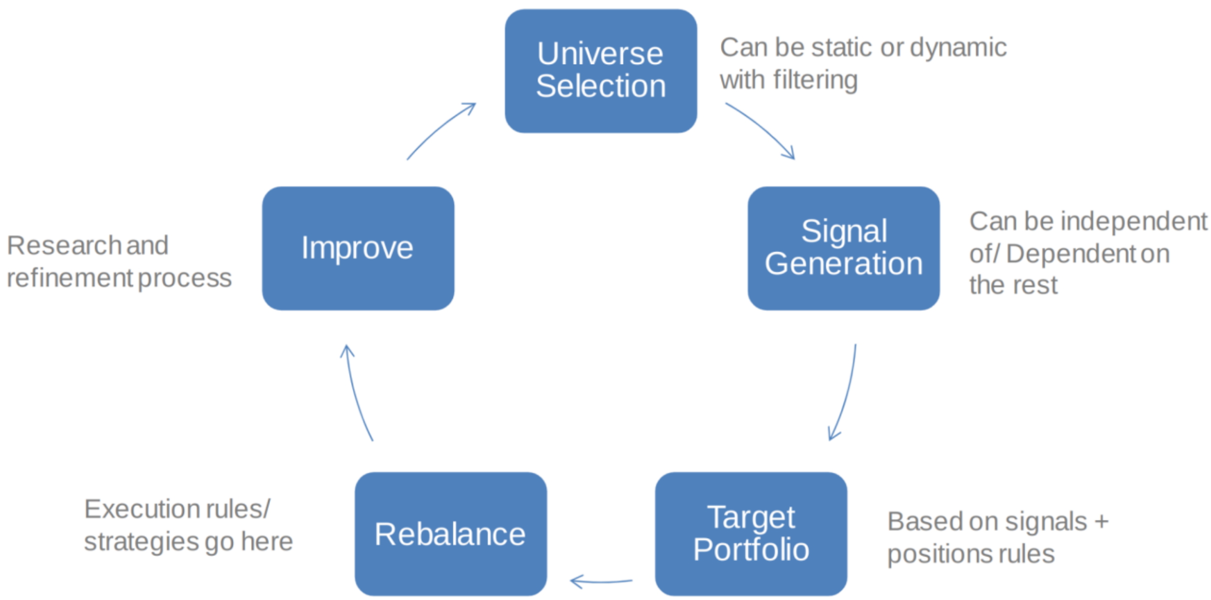
What is Blueshift?

Blueshift is a systematic trading development platform with built-in market data covering multiple markets and asset classes, powered by Zipline.

The following snapshot shows the features provided by the Blueshift platform.

Systematic strategy design cycle

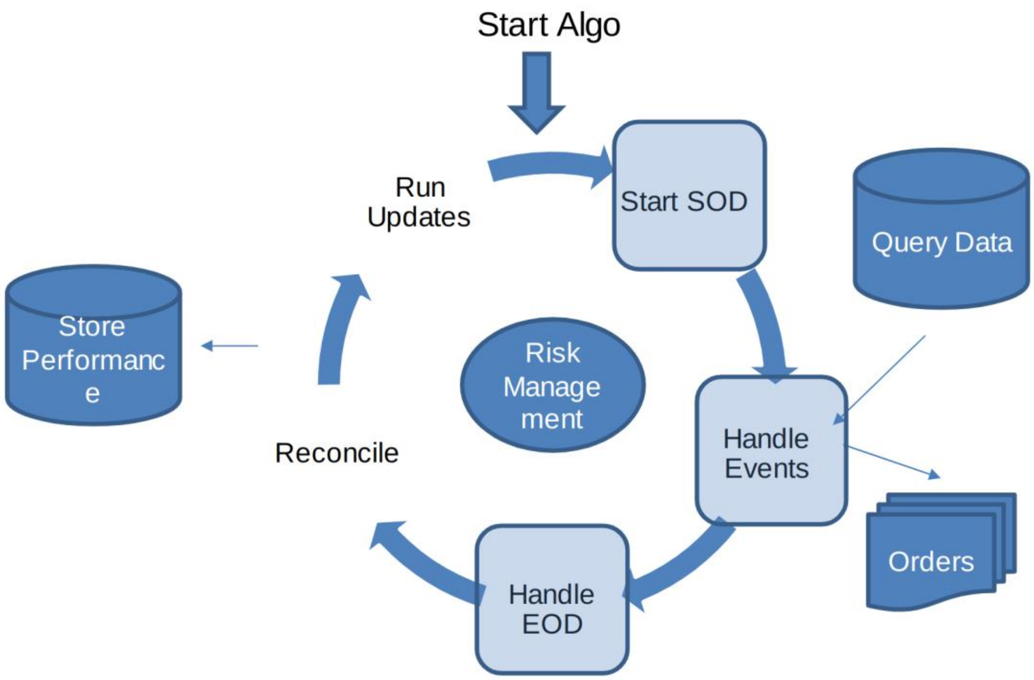
The following snapshot shows the various steps in the life cycle of a strategy.



Trading system from a user's perspective

A trading algorithm is essentially a process that consumes data and generates orders. Other things are additional but essential paraphernalia (risk and capital management, performance tracking etc.)

The following snapshot shows the design of a trading system from a user's point of view.



<u>**How to develop a systematic trading strategy?**</u>

- **Inputs: Input to the strategies can come in many flavours**

  a. Price/returns and its transformation, volumes and open interest data, macroeconomic indicators, Twitter sentiments, etc.

- **Trading Rules/Logic: Two approaches** -

  1. Form a hypothesis (e.g. moving average crossover signals change in trends) and test
  2. Feed the data to infer rules (the subject matter of ML/AI)

- **Backtesting and forward testing:**

  a. A good platform to guard against biases: data-mining bias, survivorship bias, market-impact modelling, look-ahead bias
  b. Should be flexible, event-driven (to avoid look-ahead bias) and with built-in analytics

- **Portfolio creation/optimization**: Never go all-in with a single strategy

  a. Two strategies are better than one if they are uncorrelated
  b. Various methods exist – ensemble, traditional portfolio theory, dynamic portfolio allocation, etc.

Blueshift workflow

Datasets on Blueshift

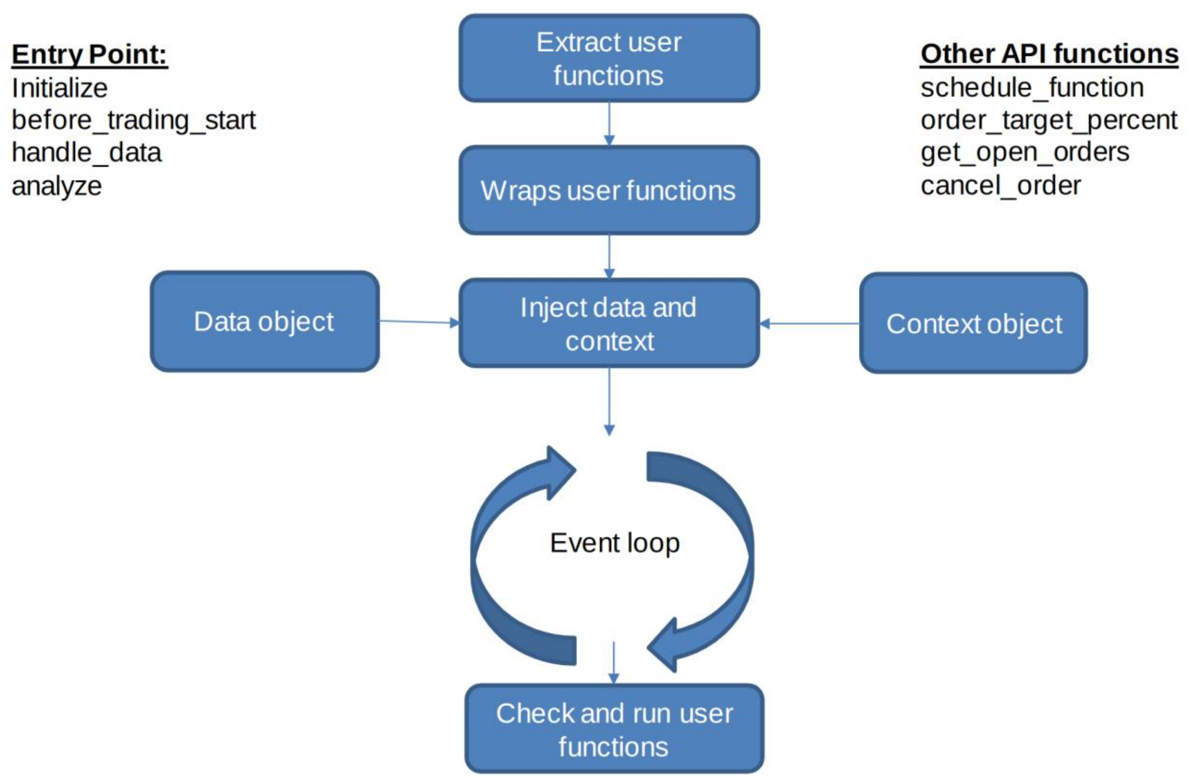
- Equities market data for NYSE (the US) and NSE (India)
  - Minute levels with corporate actions.
  - It is updated once every day after the market closes.

- FX data
  - 10 currency pairs - AUD/USD,EUR/CHF, EUR/JPY, EUR/USD, GBP/JPY, GBP/USD, NZD/USD, USD/CAD, USD/CHF, USD/JPY.
  - Minute-level data since 2008.
  - It is updated once every day.

- Macro data on eight underlying economies (real GDP, Y-o-Y core inflation rate, short term (3-month) and long term (10-year) interest rates.

How does Blueshift's API work?

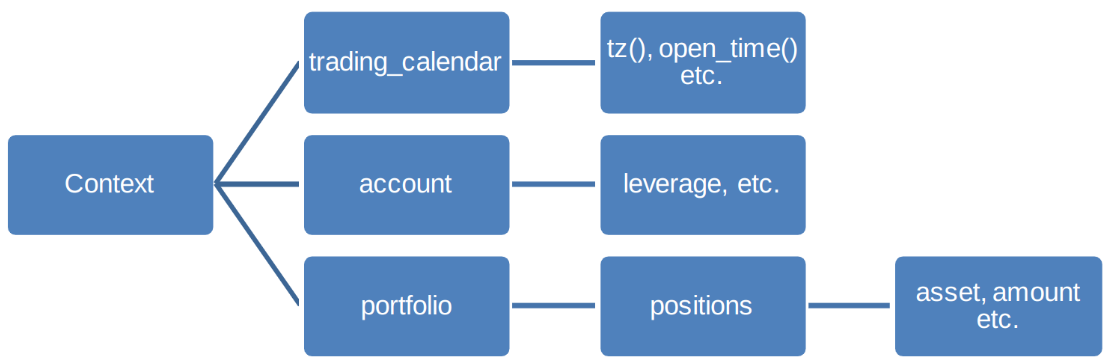
The following snapshot shows the workflow and the functions involved in the API execution process.



Blueshift environment

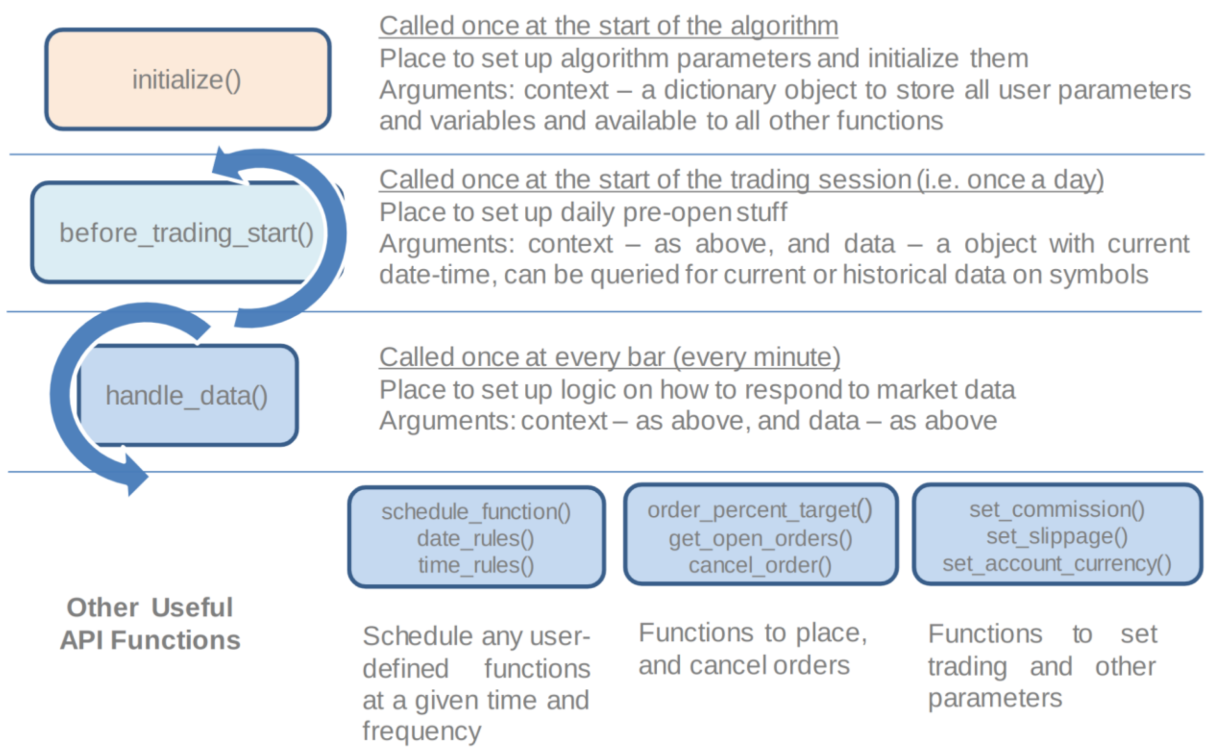
- Data contains everything needed to feed data: current(), history(), as well as current_dt

- Context contains everything else: Portfolio (context.portfolio), account (context.account), calendar (trading_calendar)



Workflow on Blueshift platform

The following snapshot shows the function-wise workflow on the Blueshift platform.



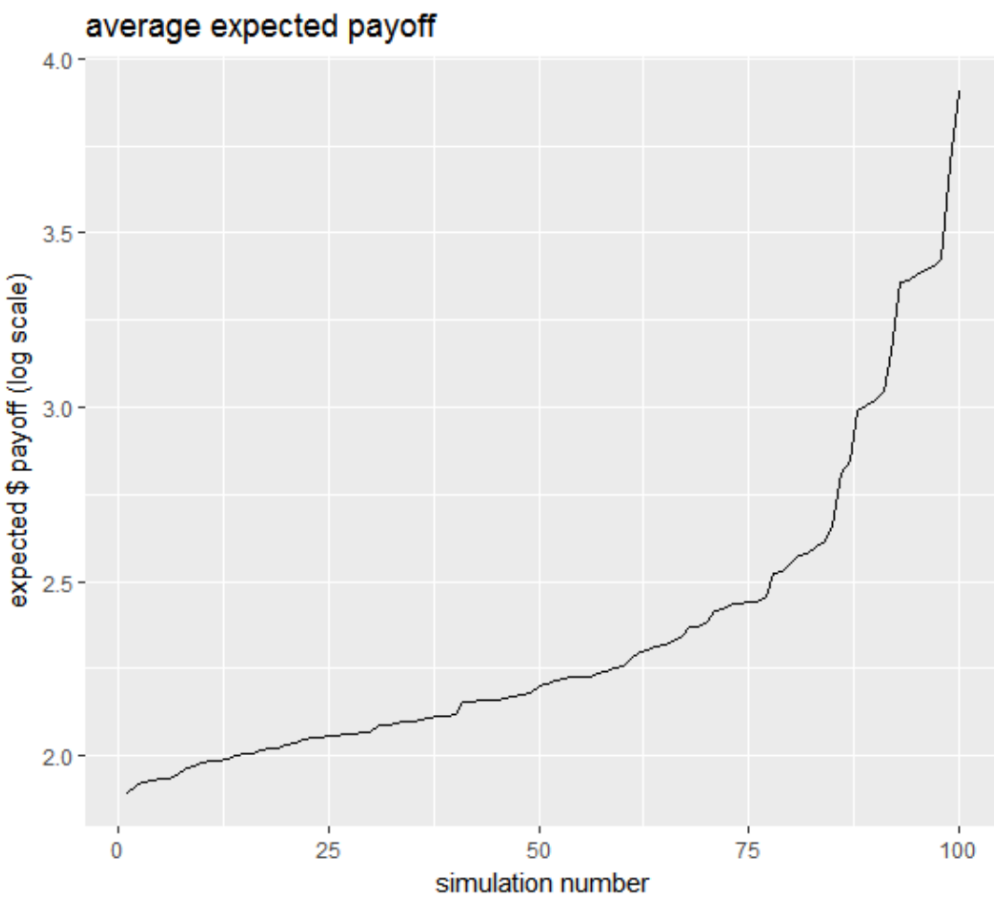
Alpha strategies: understanding the source of risks and returns

Asset pricing: basics

- Fundamental value: The intrinsic value of the asset
- Market price: The price at which players can agree to buy or sell an asset.

Asset pricing: paradox

The following snapshot shows the plot of the expected payoff for various simulations.



People can have unbounded expectations!!

Asset pricing: concepts

Utility: A mathematical concept (a function) to arrange a set of choices in a given time and over time in order of preference. Simply put, the pleasure or satisfaction that is derived.

The value of anything is the present expected utility **(Expected Utility Hypothesis)**

Present value models: Fundamental value is Net Present Value (NPV) or Discounted Cash Flow (DCF) appropriately adjusted for costs (including opportunity costs) and risks.

- Time Value of Money (TVM): People prefer now to later.
- Risk aversion: People prefer less risky options when the payoff is the same.
- Assumes functional capital market (lending and borrowing)

Risk-free rate: It captures the opportunity costs.

Risk premium: Captures risk aversion – how much investors demand to be compensated for risks in equilibrium

Discount Factors: Include both risk premia and time premia. It depends on individual utility, but in the presence of capital markets, converges to give "Law of One Price."

<u>Systematic equities: A brief overview</u>

Momentum: Exploiting time-series persistence

Strategy - I: Buy (sell) assets that have been going up (down) in prices in the past, hold them in the portfolio for a defined period (holding period) and then re-balance. This is known as **time-series momentum.**

Strategy - II: Rank all assets based on their past returns. Buy the top x-percentile and sell the bottom x-percentile (even if they are going up in prices). Re-balance after the holding period.

Systematic asset management before momentum

1. Passive - Portfolio optimization
2. Active - Market timing and signal generation (digital signal processing, technical analysis etc.)

Systematic asset management after momentum: factor investing

1. Passive - Portfolio optimization (mean-variance)
2. Passive - Portfolio optimization (risk factors optimization)
3. Active - Market timing and signal generation (digital signal processing, technical analysis etc.)

What drives momentum?

Behaviour

Under-Reaction: Slow reactions to company-specific or macro-economic news (the opposing theory is the Efficient Market Hypothesis or EMH). Conservative behaviour, bounded rationality, liquidity constraints etc.

Delayed over-reaction: Behavioural feedback mechanism. This also gets reinforced by under-reaction.

Disposition effect: Investors tend to stick to losers too long and book profit too quickly.
It creates momentum.

Risk-based

Risk compensation: The firm's market value growing faster than the fundamental leads to an increase in risk; hence momentum profit should compensate. Momentum is usually stronger for companies with higher growth opportunities and riskier cash flows.

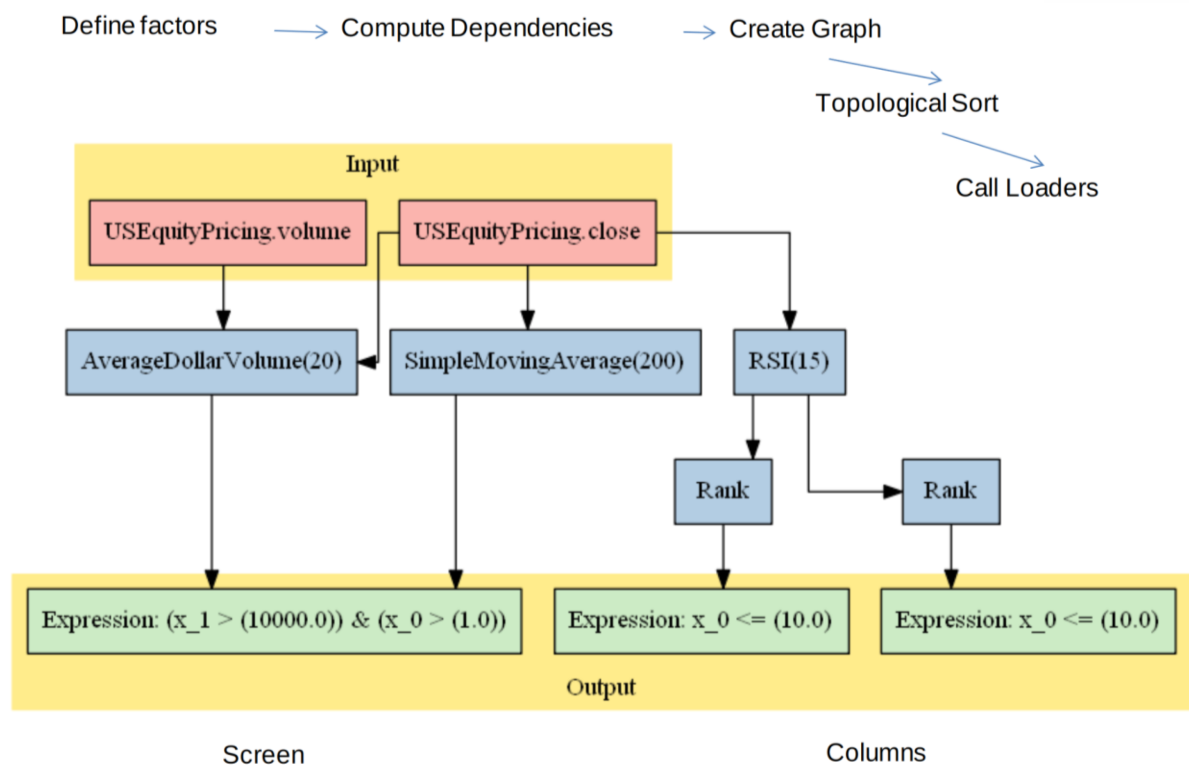
Correlation structure: Momentum assets strongly correlate with each other, which implies a common factor that can experience changes in future (momentum crash)

Liquidity risks: Recent winners have greater exposure to liquidity risks (everyone tries to exit at the same time)

Blueshift pipelines

Pipelines are efficient ways to compute a factor on a large universe.

The following snapshot shows an example of the Blueshift pipeline.



<u>Statistical arbitrage</u>

Concept

- Choose a pair of stocks that move together very closely, based on certain criteria (i.e. TCS & Infosys)
- Wait until the prices diverge beyond a certain threshold, then short the “winner” and buy the “loser” (contrarian approach)
- Reverse your positions when the two prices converge --> Profit from the reversal in trend
- Pairs trading strategy requires a few key decision points:
    - Pair (or basket) selection algorithm
    - Market timing (entry and exist)
    - Optimal position sizing mechanism
    - Spotting the scarce opportunity

Risks

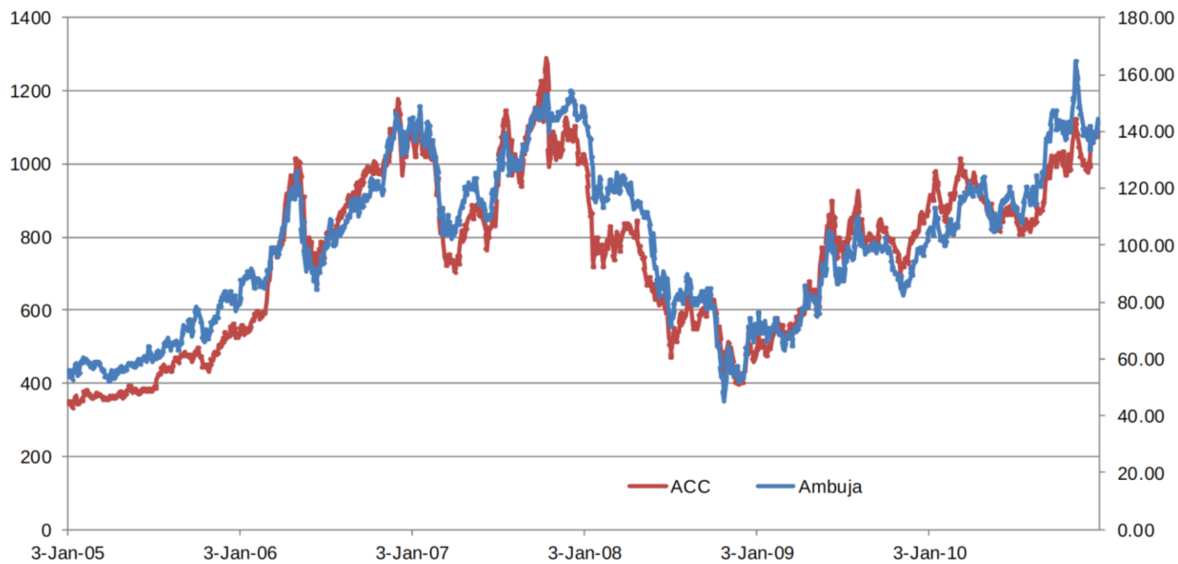
Statistical arbitrage is not a risk-free strategy. The critical risk involved in trading are as follows:

- Rather than converging, prices of the two securities can begin to diverge (drift apart), i.e. the spread picks uptrend rather than mean-reverting to the original mean, and the cointegration can be broken.

- Sometimes, an event in security can trigger extreme movement in pair ratio. Entering into such a security trade should be avoided.
- Strict risk management of stop-loss is required to handle adverse situations once the mean-reverting behaviour is invalidated.
- Inherently statistical arbitrage is like a short gamma trades. They tend to win most of the time and sometimes blow up.

Example -

The following snapshot shows the example of a pair (ACC Ambuja). Observe how closely they follow the same trend.



Cointegration

If there is a stationary linear combination of non-stationary asset prices, the combined prices are cointegrated. Stocks in the same industry and similar business models tend to be cointegrated. We can determine, statistically, if two series are cointegrated or not by using:

- Augmented Dickey-Fuller (ADF) test
- Johansen test

Steps for statistical arbitrage

- Pair identification
  - Based on prior knowledge
  - Based on quantitative analysis (example: cluster analysis)
  - Finally, confirm the hypothesis by cointegration test

- Rich cheap identification?
  - Track the cointegrated pair and compute z-spread
  - Define threshold of entry and exit

        - Check for idiosyncratic nature (news flows, company-specific announcements etc.)

- Risk management
    - Position sizing: based on the expected time to convergence, spread volatility over the period and strength of the signal
    - Step in vs all-in sizing

Systematic trading in FX

The fundamental beta in FX

- Value
- Momentum
- Carry
- Defensive

How to create strategies?

1. Which one to buy and which one to sell?
2. Code the strategy
3. Add the momentum logic
4. Adjust the re-balance function
5. Run the backtest.

Backtest evaluation

Risk reward metrics

Algorithm metrics:
Risks and performance: Total returns, annualized returns, volatility, Sharpe ratio, Sortino ratio, Omega ratio, time-series stability, skew, kurtosis, max drawdowns, alpha, beta, net and gross leverage.

Trade metrics:
The average number of trades, per trade profit, win rate, total trades, profit-loss ratio, average wins, average loss, average holding period, average costs, maximum adverse excursion, maximum favourable excursion.

Scenario analysis:
Strategy performance in defined periods – 2007 quant crisis, 2008 financial crisis, 2011 US Budget crisis, 2010-2012 European sovereign crisis, EM crisis, Other identifiable periods of idiosyncratic risks.

The following snapshot shows the most commonly used ratios to understand the risk-reward behaviour of a strategy.

Sharpe Ratio:

$$Sharpe = \frac{R_P - R_f}{\sigma_P}$$

Sortino Ratio:

$$Sortino = \frac{R_P - R_B}{\sigma_{down}}$$

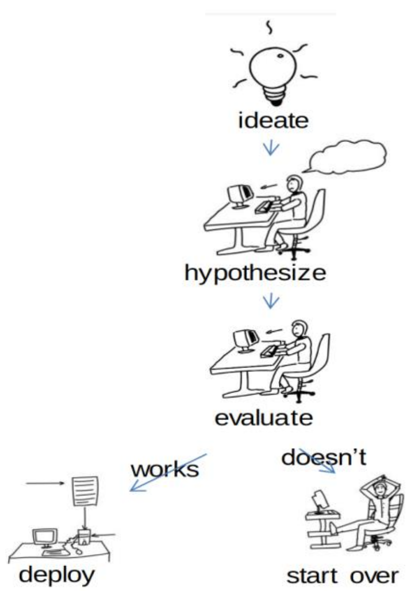
Omega Ratio:

$$\Omega = \frac{\int_r^{\infty}(1 - F(x)).\,dx}{\int_{-\infty}^{r} F(x).\,dx}$$

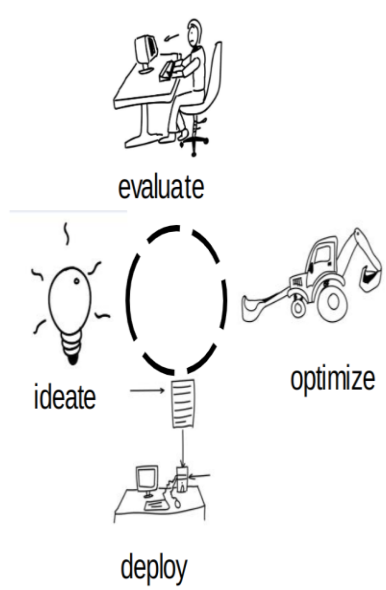
Is it a good idea to optimize backtest strategies?

The following snapshot shows the scientific way of developing a strategy compared to how most of us end up doing it.

Scientific way of developing a strategy:

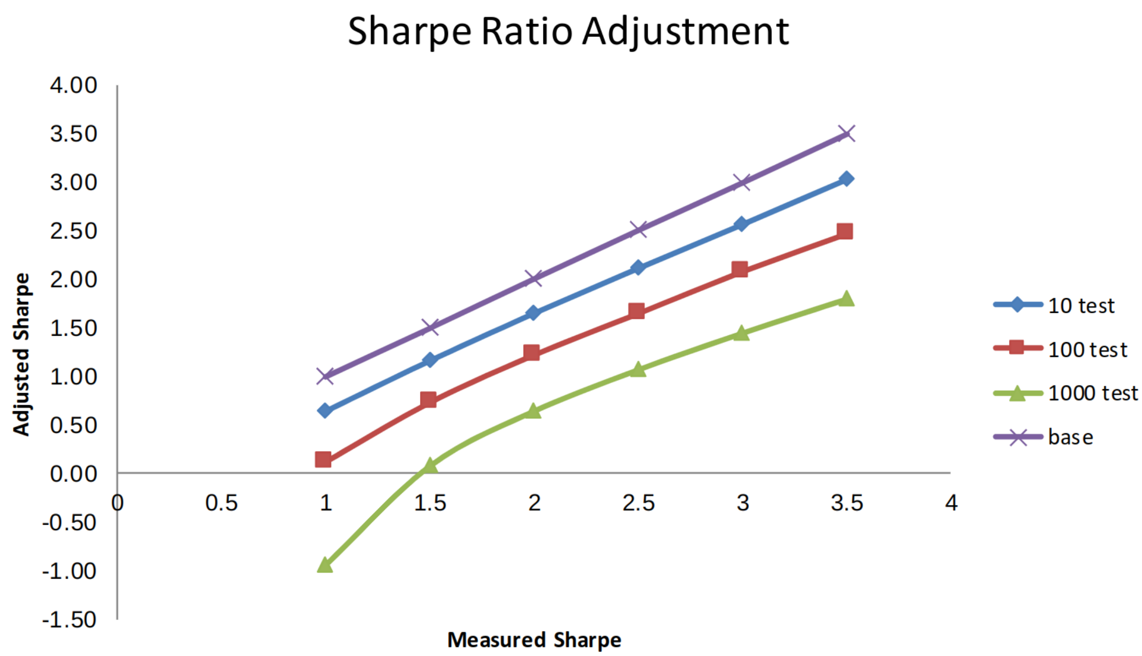


The way most of us end up with:



Problems with optimization

The following snapshot shows the adjusted Sharpe ratio for multiple no. of tests



Alternative strategies

1. Adaptive strategies: Example change point analysis
2. Stable factor research: With economic/behavioural justification
3. Ensemble strategies: Works best with diverse strategies
4. Validation: cross-validation or paper-trading

Systematic strategy: What it is not?

- **Black box** - A strategy shouldn't be a black box. We should always explain why we buy what we buy and why our strategy should make money.
- **Risk-free profit** - No systematic strategy is risk-free. Risk management is crucial.
- **Fire and forget** - A systematic strategy must pass through a constant cycle of the performance measure, tuning and re-risking.
- **Man vs Machine** - Systematic strategies are not about man vs machine; they're about man and machine. Human beings are good at developing hypotheses, and machines are good at testing and executing them.

Recommendations for building strategies

Here are some recommendations while developing strategies -

- Save strategy variables in the ‘context’ environment - It helps in tracking variables
- Use schedule_function() to run faster code
- Keep a check on account leverage
- Don’t overfit - Build a hypothesis and test
- Check the statistics in the backtest results -
    - A strategy with a low Sharpe ratio,
    - Concentrated wins (low stability),
    - Large downside risks (negatively skewed),
    - High beta,

are not something you would want to invest in.

List of demo strategies

Here’s a list of demo strategies that can be found on Blueshift’s GitHub page -

1. Strategy template
2. Value strategy
3. Carry strategy
4. Time-series momentum strategy
5. Cross-sectional momentum strategy
6. Factors basket strategy
7. Bollinger band based break-out strategy
8. RSI based mean-reversion strategy
9. Correlation-based statistical strategy
10. FX daily computation template
11. Change point-based regime switching strategy